Network Design & Bill of Materials (BOM)

Project Scope: High-Capacity 6-Zone Point-to-Multipoint (PtMP) Wireless Network for 48 PTZ Streaming Cameras

Executive Summary & Bandwidth Requirements

Connecting 6 nearby zones with 8 PTZ streaming cameras per zone (48 cameras total) creates an intensive network traffic environment. Because video streaming demands continuous, high-volume upload capabilities, standard low-cost wireless setups are prone to failure. This document consolidates two structural paths alongside vital deployment precautions.

The Bandwidth Calculations:

- **Per Camera Load:** 1x HD/2K PTZ Camera requires ~6 to 8 Mbps upload.
- **Per Zone Load:** 8x Cameras per zone = ~48 to 64 Mbps continuous load.
- **Total Network Load:** 6 zones × 64 Mbps = **~384 Mbps** continuous, non-stop upload bandwidth hitting the central base station.

Option A: The Gold Standard (60GHz Ubiquiti Wave)

Operating in the 60GHz spectrum, this option bypasses crowded 5GHz frequencies completely. It provides fiber-equivalent multi-gigabit throughput and is immune to typical local wireless interference. Ideal for instantaneous PTZ control response.

Qty	Item Description	Est. Unit (RM)	Est. Total (RM)
2	Ubiquiti Wave AP / Wave AP Micro (60GHz Base Stations)	2,600	5,200
6	Ubiquiti Wave Nano (60GHz Station Receivers)	950	5,700
6	Ubiquiti Ultra 60W PoE Switch (USW-Ultra-60W)	750	4,500
6	Ubiquiti PoE+ Injectors (U-POE-at)	75	450
2	Outdoor Shielded Cat6 Cable (305m Drum)	550	1,100
8	Universal Outdoor J-Pole Mounts	80	640
	Estimated Option A Total		**RM 17,590**

Option B: The Segmented Route (5GHz Ubiquiti AirMax)

A budget-friendly approach that uses two separate 5GHz Base Stations matching sector antennas to distribute the massive 384 Mbps camera load onto independent wireless channels. This prevents single-frequency saturation.

Qty	Item Description	Est. Unit (RM)	Est. Total (RM)
2	Ubiquiti Rocket 5AC Lite (5GHz Base Station Radios)	750	1,500
2	Ubiquiti AirMax 5GHz 60° Sector Antennas	900	1,800
6	Ubiquiti NanoBeam 5AC (High-Gain Station Receivers)	550	3,300
6	8-Port Gigabit PoE+ Managed Switch (TP-Link Omada - 150W)	620	3,720
2	Outdoor Shielded Cat6 Cable (305m Drum)	550	1,100
8	Universal Outdoor J-Pole Mounts	80	640
	Estimated Option B Total		**RM 12,060**

Critical Architectural & Deployment Points

1. Power Budgets vs. PTZ Startup Spikes

Unlike static security cameras which pull a stable 5-7 Watts, PTZ cameras use motorized mechanical gears to pan, tilt, and zoom. When an entire zone reboots due to a power trip, all 8 PTZ cameras perform a startup calibration sweep simultaneously. This mechanical activity can spike the power consumption to 20-30W per unit. Ensure your PoE switch power budget handles total peak wattage, not just idle wattage.

2. Environmental Hardening (Weatherproofing)

Because these local switches and power accessories are located outdoors at individual zones, they must be enclosed in IP66-rated weatherproof boxes. Condensation and extreme heat will prematurely degrade enterprise electronic components; using passive cooling vents or outdoor sun-shields on your enclosures is highly recommended.

3. Cabling Infrastructure & Grounding

Standard indoor PVC Ethernet lines will crack under UV exposure within months, letting rain seep in to fry your switches. You must use Outdoor-Rated Shielded Cat6 Cables. Additionally, ensure proper grounding using Ubiquiti Ethernet Surge Protectors (ETH-SP-G2) at the point where the cable drops down from the roof mast into the zone box to avoid lightning surge destruction.

4. Unobstructed Line of Sight (LoS)

For 60GHz equipment especially, clear Line of Sight is absolute. Even a single growing tree branch blocking the visual path can break the link entirely. If dense obstacles exist, the 5GHz configuration is more forgiving but still requires clear visual clearance across the top 60% of the wireless Fresnel Zone.

5. Staging and Bench Configuration

Never configure these items up on a roof or pole. Set up all 1 Base Station, 6 Receivers, 6 Switches, and test cameras in a single office room first. Assign static IP addresses (e.g., Base Station: 192.168.1.20, Client 1: 192.168.1.21, Client 2: 192.168.1.22, etc.). Label every single device with a physical sticker before deployment.